Lab 2 · Quality Control and Genome Assembly

Day 1 · 13:45–15:30 · 105 min · Pairs (half the room does the Illumina track, half the Nanopore track) Objective: LO3 — Perform QC and assembly on raw reads in Galaxy and judge whether an assembly is good enough to use

> **The isolate:** an MRSA strain (KUN1163) sequenced **twice**, once on Illumina and once on Oxford Nanopore. At the end, an Illumina pair and a Nanopore pair compare results.

> ⏱️ **Strategy:** assembly takes **20–60 minutes** on Galaxy. We **start the assembly early** and interpret QC while it runs. If it hasn't finished by 15:00, import the instructor's pre-computed result (Part E).

Part A — Set up (10 min)

1. Create a new history called `Lab2 – Illumina` **or** `Lab2 – Nanopore`
2. Click **Upload** → **Paste/Fetch data** and paste the URL(s) for your track:

Illumina track (paste both lines at once):

```
https://zenodo.org/record/10669812/files/DRR187559_1.fastqsanger.bz2
https://zenodo.org/record/10669812/files/DRR187559_2.fastqsanger.bz2
```

Set **Type** → `fastqsanger.bz2`

Nanopore track:

```
https://zenodo.org/record/10669812/files/DRR187567.fastq.bz2
```

Set **Type** → `fastqsanger.bz2`

3. Click **Start**. Galaxy downloads the files directly from the internet, so your laptop does **not** need to download 500 MB.
4. Rename the datasets: `illumina_R1`, `illumina_R2` **or** `nanopore_reads`

✍️ **Check yourself A:** Why are there two Illumina files but only one Nanopore file?

ILLUMINA TRACK

Part B-I — Quality check (15 min)

1. Tool **`FastQC`** → input: `illumina_R1` → **Run**. Repeat for `illumina_R2`.
2. Open the **Webpage** output and look at:
   - **Basic statistics:** total sequences and sequence length
   - **Per base sequence quality:** is most of the box plot in the green zone?

- **Adapter content:** are adapters present?

Part C-I — Clean the reads (10 min)

1. Tool **`fastp`**
   - "Single-end or paired reads" → **Paired**
   - Input 1 → `illumina_R1`, Input 2 → `illumina_R2`
   - Leave the other settings at their defaults (fastp detects adapters automatically and trims low-quality ends)
2. **Run**. Outputs: two trimmed read files and an HTML report.
   - *If Shovill doesn't offer the fastp outputs as inputs:* click the pencil ✏️ on each fastp read output → **Datatypes** → set it to `fastqsanger.gz` → **Save**
3. Open the HTML report: what **percentage of reads passed** the filters?

Part D-I — Assemble (start now and let it run)

1. Tool **`Shovill`**
   - "Input reads type" → **Paired-end**
   - Forward → fastp read 1 output; Reverse → fastp read 2 output
   - Leave the defaults
2. **Run**. It will take a while, so **move on to Part F (interpreting QC) while it runs**.

NANOPORE TRACK

Part B-N — Quality check (15 min)

1. Tool **`NanoPlot`** → input: `nanopore_reads` → **Run**
2. Open the **HTML report** and find:
   - **Number of reads**
   - **Read length N50** (half of all bases are in reads at least this long)
   - **Mean read quality**
   - The **read length vs quality** plot

Part C-N — Filter the reads (10 min)

1. Tool **`filtlong`**
   - Input → `nanopore_reads`
   - "Min. length" → `1000` (reads shorter than 1,000 bases make assembly harder)
   - Leave the other settings at their defaults
2. **Run**

Part D-N — Assemble (start now and let it run)

1. Tool **`Flye`**
   - Input reads → filtlong output
   - "Mode" → **Nanopore corrected** (`--nano-corr`), as in the GTN tutorial for this dataset
   - Leave the defaults
2. **Run**. It will take a while, so **move on to Part F while it runs**.

BOTH TRACKS

Part F — Interpret QC while the assembly runs (20 min, discussion within your pair)
Fill in this table:

	Your track
Number of reads	
Typical read length	
Total bases (≈ reads × length)	
Expected genome size (*S. aureus*)	≈ 2.8 million bases
Estimated coverage (total bases ÷ genome size)	
Is the quality good enough? (Y/N + why)	

✍️ **Check yourself F:**

1. Is your coverage above the minimum recommended (≈30× Illumina, ≈40–50× Nanopore)?
2. The instructor will show a FastQC report from a *bad* run. What looks different?

Part G — Assess the assembly with QUAST (15 min)
When your assembly dataset turns green:

1. Tool **`Quast`**
   - Contigs/scaffolds → your Shovill `contigs` **or** Flye `assembly` output
   - Leave the defaults (no reference)
2. **Run**, then open the **HTML report**

Metric	What it means	Good sign for *S. aureus*	Your value
# contigs	Number of pieces	Illumina: tens to ~100 · Nanopore: 1–5	
Total length	Sum of all contigs	≈ 2.7–3.0 Mb	
Largest contig	Longest piece	Nanopore: ≈ whole chromosome	
N50	Half the genome is in contigs at least this long	Bigger is better	
GC (%)	Proportion of G+C	≈ 33% for *S. aureus*	

✍️ **Check yourself G:**

1. Is the total length close to the expected genome size? What would a total length of **5.6 Mb** suggest? *(Hint: two organisms?)*

2. Would you accept this assembly for AMR analysis? Why?

Part H — Compare tracks (10 min, Illumina pair + Nanopore pair)
Pair up with a team from the other track and compare QUAST results:

	Illumina (Shovill)	Nanopore (Flye)
# contigs		
Largest contig		
Total length		
N50		

Discuss:

- Which assembly is more **complete** (fewer pieces)?
- Which is likely more **accurate** at single-letter level? *(See L2.)*
- If you needed to know whether a resistance gene sits on a **plasmid**, which would you prefer?
- What would a **hybrid** assembly (both read types) give you?

Part E — Fallback: import pre-computed results

If your assembly is still running at 15:00:

1. Open the shared history (the link is also on the board): https://usegalaxy.eu/u/aryakaul/h/ughe-ssi-course-precomputed
2. Click **Import**. A copy of `UGHE-SSI course - precomputed` opens as your current history
3. Use `Lab2_Illumina_Shovill_contigs.fasta` **or** `Lab2_Nanopore_Flye_assembly` and its QUAST report
4. Continue from Part G. **Your own job will keep running** in your Lab 2 history. Check it tonight!

Summary

```
Illumina:  reads → FastQC → fastp → Shovill → QUAST → many accurate contigs
Nanopore:  reads → NanoPlot → filtlong → Flye  → QUAST → few long contigs (≈
complete genome)
```

Your Galaxy history now holds a complete, reproducible record of how you went from raw reads to a genome. Tomorrow we ask: **what is in this genome?**

Extension (fast pairs): try `Unicycler` hybrid assembly with the Illumina and Nanopore reads together, or run `Bandage` on the assembly graph to *see* the genome.

Adapted from the Galaxy Training Network tutorials "Bacterial Genome Assembly using Shovill" and "Bacterial Genome Assembly using Flye" (CC-BY 4.0), training.galaxyproject.org/training-material/topics/assembly/